



15
Years
Anniversary

How Local Incentives Shape Conflict Network Formation: An Agent-Based Study of the Myanmar Civil War

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Introduction

- Armed conflicts frequently persist even when they generate collective welfare losses.
- A central research question is why conflict actors continue to form **stable coalition structures** that are socially inefficient and which factors most strongly influence conflict persistence? E.g. natural resources, territorial control, human resources etc.
- Traditional analytical models of network formation rely on restrictive assumptions, such as homogeneity of agents, perfect and globally observable information.
- These assumptions make it difficult to study realistic conflict environments characterized by **heterogeneity, local incentives, and incomplete information**.
- This project studies the emergence of conflict networks using an **agent-based modeling (ABM)** framework that allows network structures to emerge from decentralized interactions, with particular focus on the Myanmar Civil War.

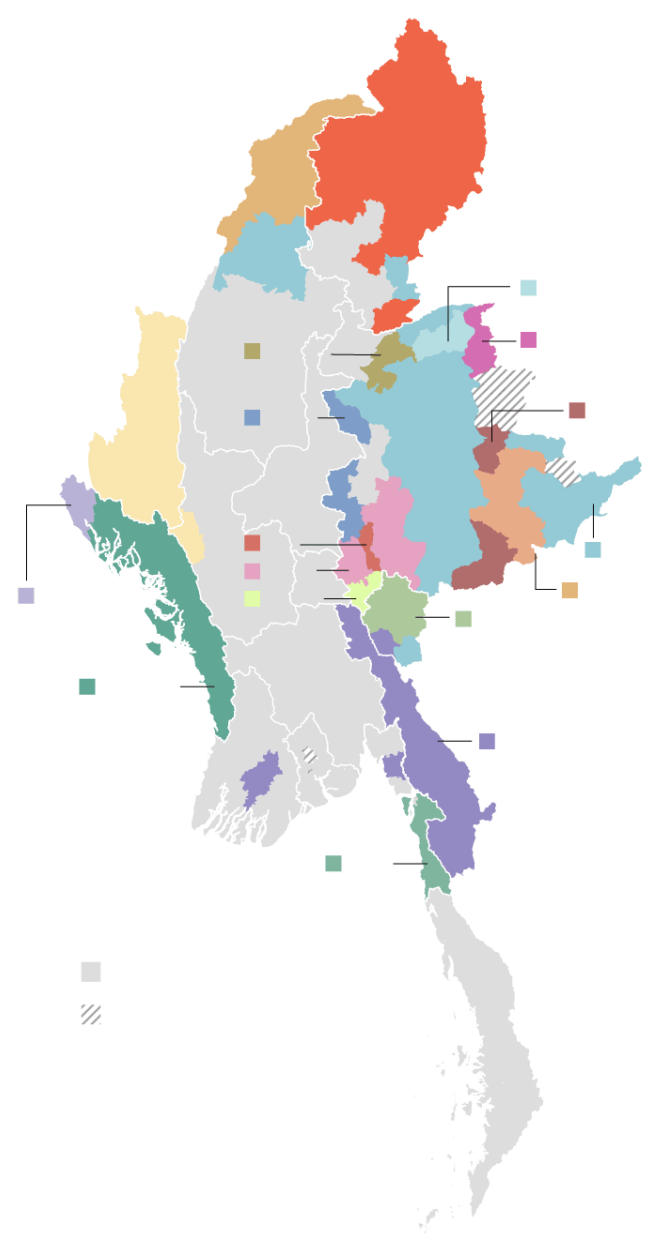


Figure 1. Map of Myanmar Civil War (Beech and Cai [2024])

Theoretical Framework

We model conflict as a **dynamic network formation** process that arises from decentralized interactions. Let the set of agents be $N = \{1, \dots, n\}$, and let $g \in G$ represent a network of relationships between them. Each pair of agents $i, j \in N$ can have three states: Friendship, Enmity, and Neutrality.

Such a network can be represented by an *adjacency matrix* of size $n \times n$, where

$$g_{ij} = \begin{cases} 1 & \text{if } i \neq j \text{ and } i \text{ and } j \text{ are allies} \\ -1 & \text{if } i \neq j \text{ and } i \text{ and } j \text{ are enemies} \\ 0 & \text{if } i \neq j \text{ and } i \text{ and } j \text{ are neutral, OR } i = j \end{cases}$$

Each agent i has utility function u_i which depends on agent's resources, position in the network and location on the grid. We represent agents as decision-making units that aim to maximize their utility. Agents have incomplete information about the system. As a result, they choose actions based only on their local observations, which ensure decentralized interaction.

The system evolves over discrete time-steps t . At each step, the state of the system changes as agents interact locally. We study how such local interactions generate global patterns in the evolution of the network.

Applied Simulation Design

- We start with approximating the shape of Myanmar using a square grid and assign each tile of the grid a resource endowment that corresponds to the real distribution of resources in that area.

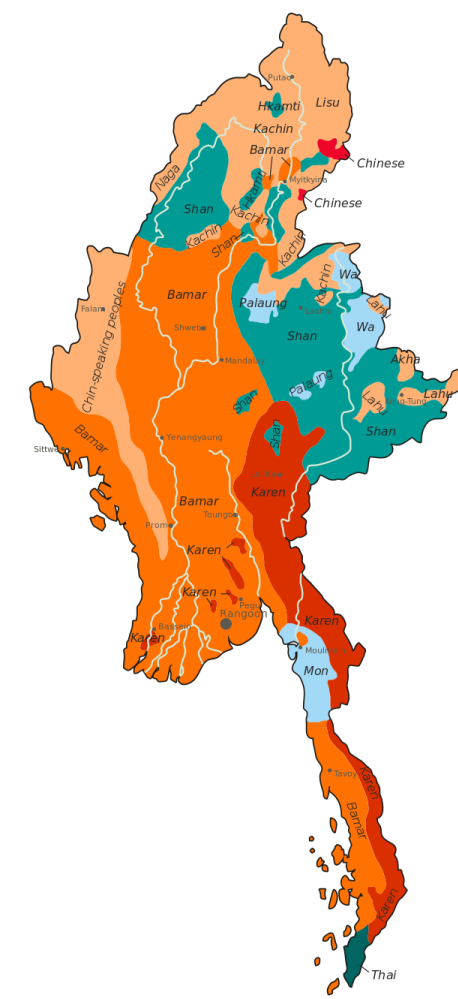


Figure 2. Map of Myanmar (Adrienne et al.)

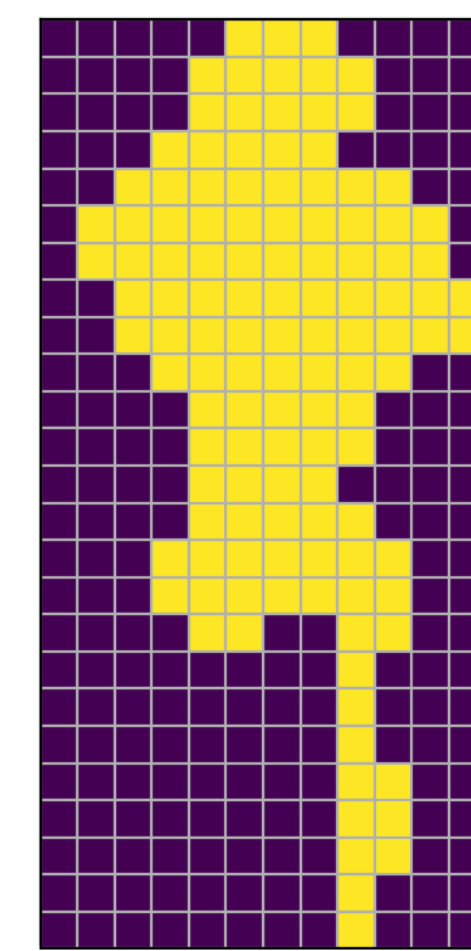


Figure 3. Map of Myanmar, grid approximation

- A subset of key actors is placed on the grid according to their geographical location.
- By using data from ACLED, we form an initial network of alliances and enmities.
 - If two actors have fought against each other at least once and never fought on the same side, they are connected with the enmity edge.
 - If two actors have fought together at least once and never opposed each other, they are connected with a friendship edge.
- Once model starts:
 - Each agent selects a tile within its vision radius that maximizes its utility.
 - If tile is not contested, the agent occupies it and shares resources with its allies.
 - If multiple agents contest a tile, a conflict occurs, and cost depends on their resource and alliance network.
- Alliance and rivalries evolve in decentralized manner:
 - Forming an alliance reduces the cost of the conflict but requires sharing resources
 - Conflicts generate enmity links, which may later dissolve if cooperation becomes beneficial.

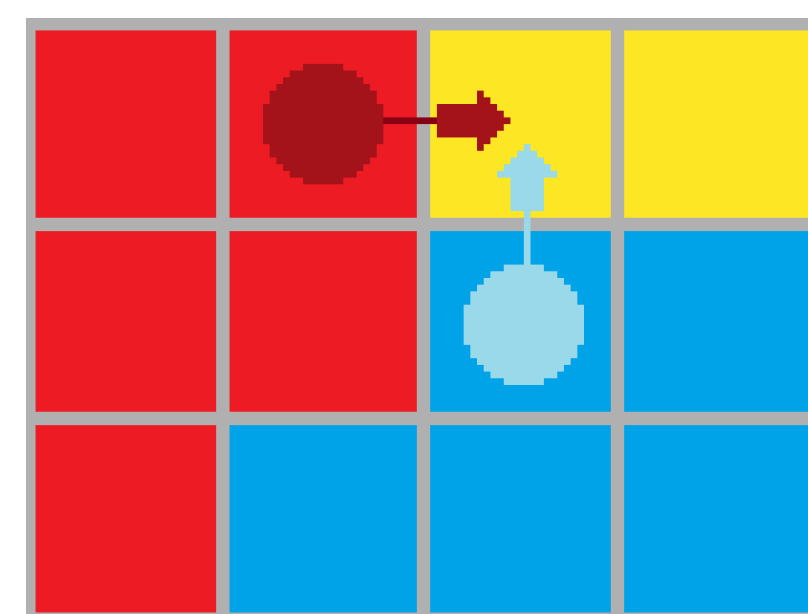


Figure 4. Agents contesting for a tile

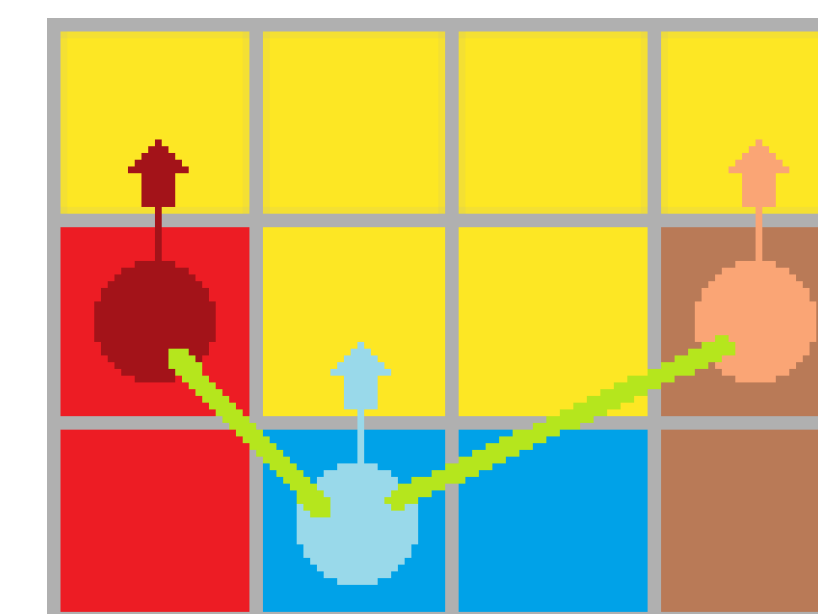


Figure 5. Agent A (blue) occupying free tile while in alliance with other two

- Agents have limited information about each other—it introduces uncertainty in the conflict and affects alliance decisions
- Simulations under different parameters generate evolving conflict networks that can be compared with empirical patterns in the real-world conflicts.

Data

- To construct initial conflict network, we use data from **ACLED dataset** (Armed Conflict Location and Event Data).
- Event-level data include:
 - Actors involved in conflict events
 - Geographic location
 - Time of the events
 - Type of the events (e.g. Battle, Strategic development)
- The analysis focuses primarily on the **Myanmar Civil War**.
- The dataset allows to reconstruct evolving relationship between actors across time.

Preliminary results

- Using ACLED data, we constructed the conflict network of the Myanmar Civil War with over 30 active actors.

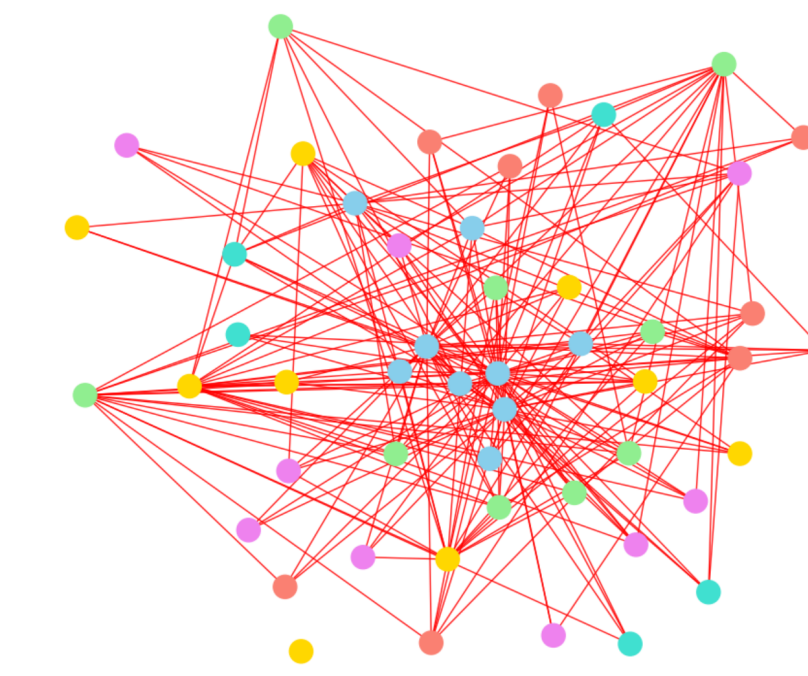


Figure 6. Enmity network of the Myanmar Civil War (2021 - Present)

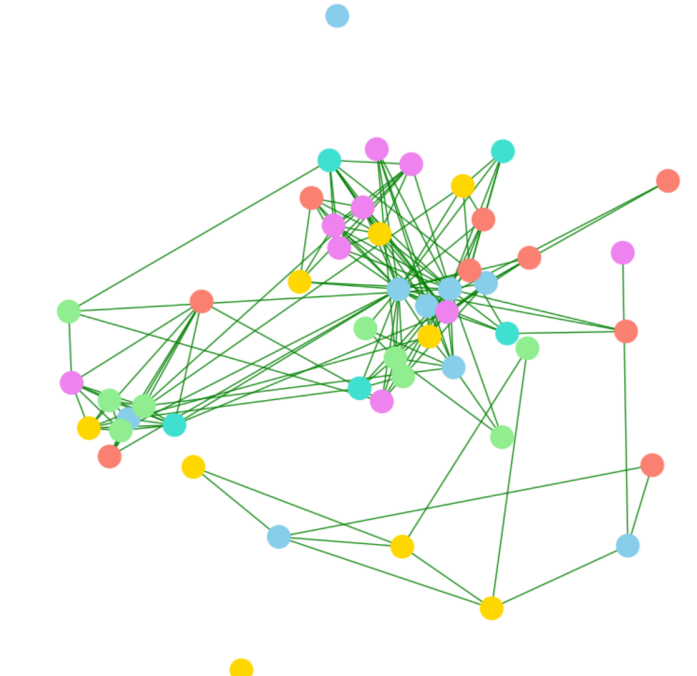


Figure 7. Alliance network of the Myanmar Civil War (2021 - Present)

- In previous work on the War in Darfur in Sudan, we constructed the conflict network and performed centrality analysis of the network to identify the most important participants—it provided an alternative ranking based on participants' network position rather than resource endowment.
- We performed agent-based simulations using distance-based utility function. By varying the parameters of the utility function and vision radius of the agents, we compared the extent to which our model converged to network topologies predicted by analytical solutions.
- Currently, our work focuses on two directions:
 - Econometric analysis of how armed conflicts affect resource expenditure among competing actors.
 - Implementation of spatial discretisation methods for grid-based setup.

Implications

- This framework allows us to study the evolution of the network by varying parameters of the model and compare it with real-world conflict dynamics.
- In particular, we can investigate:
 - Which agents, despite limited resources, significantly contributed to kickstarting the conflict.
 - Which types of resources most strongly influenced the evolution of the conflict network, and why.
 - Which actors are likely to align or compete in the future, and what incentives contribute to these decisions.
- In addition, this framework can helps us to:
 - Identify regions that require targeted policy attention by analyzing which areas in the model are most frequently contested
 - Identify which incentives could be used to reduce the persistence of conflict networks.

References

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